Lecture 22: Summary of Curve Sketching

Goal: Use the tools of calculus to draw a more accurate graph of a function.
All examples come from §4.5 of James Stewart's *Calculus Early Transcendentals*, 5th Edition as well as §4.4 of *Calculus Early Transcendentals*, 3rd Edition by Briggs et. al. (I'm too lazy to make my own examples).

All graphs will be available on Desmos (here).

Guidelines for Sketching a Curve

For drawing the graph of $y = f(x)$ by hand, consider:

(A) **Domain:** Find all values where f is defined.

(B) **Intercepts:** Find the y-intercept by computing $f(0)$. Find the x-intercepts by solving the equation $f(x) = 0$.

(C) **Symmetry:**

  (i) *Even functions:* When $f(x) = f(-x)$; this means the graph is symmetric about the y-axis.

  (ii) *Odd functions:* When $f(-x) = -f(x)$; this means it is symmetric about the origin (the left side is a 180° rotation of the right side)

  (iii) *Periodic functions:* When $f(x+p) = f(x)$ for some $p > 0$, the smallest such p is called the period. You only need to focus on one period of the function.

(D) **Asymptotes and End Behavior:**

  (i) *Vertical Asymptotes:* Occurs at $x = a$ when either:

$$\lim_{x\to a^-} f(x) \text{ OR } \lim_{x\to a^+} f(x)$$

  is not finite (equals ∞ or $-\infty$).

  (ii) *Horizontal Asymptotes:* Occurs at $y = L$ when either:

$$\lim_{x\to -\infty} f(x) = L \quad \text{OR} \quad \lim_{x\to \infty} f(x) = L$$

  (iii) *Slant Asymptotes:* The line $y = mx + b$ is a slant asymptote when either :

$$\lim_{x\to -\infty} [f(x) - (mx + b)] = 0 \quad \text{OR} \quad \lim_{x\to \infty} [f(x) - (mx + b)] = 0$$

  For rational functions, if the quotient from long division $q(x)$ is linear, then $q(x)$ is the slant asymptote.

(E) **Intervals of Increase or Decrease:** Use the I/D Test to see when f is increasing or decreasing. If $f' > 0$, then f is increasing. If $f' < 0$, f is decreasing.

(F) **Relative Maximum and Minimum Values:** Use the 1st or 2nd derivative test to determine if critical numbers are relative extrema.

(G) **Concavity and Inflection Points:** Use the concavity test. If $f'' > 0$, then f is concave upwards. If $f'' < 0$, f is concave downwards.

(H) **If needed, get more information:** You can also compute $(x, f(x))$ pairs to see the height of the graph. The tangent line also tells you where the graph is going at that point.